Cell communication and cell signaling

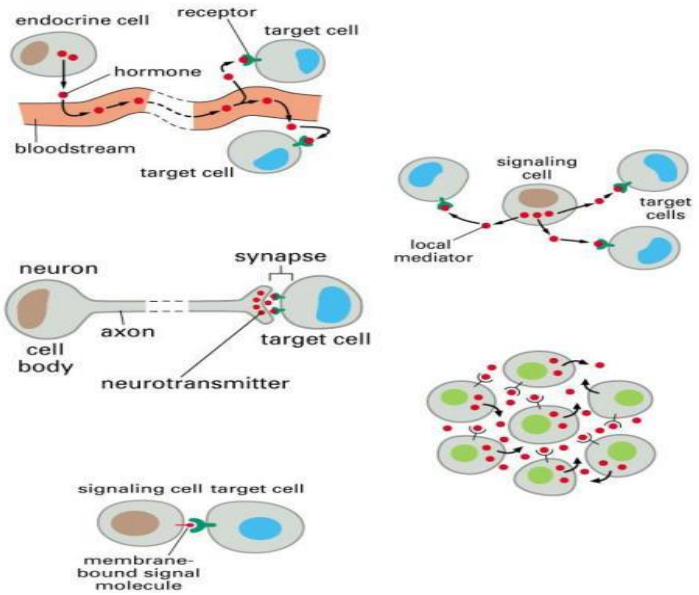
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Basic categories of chemical signalling

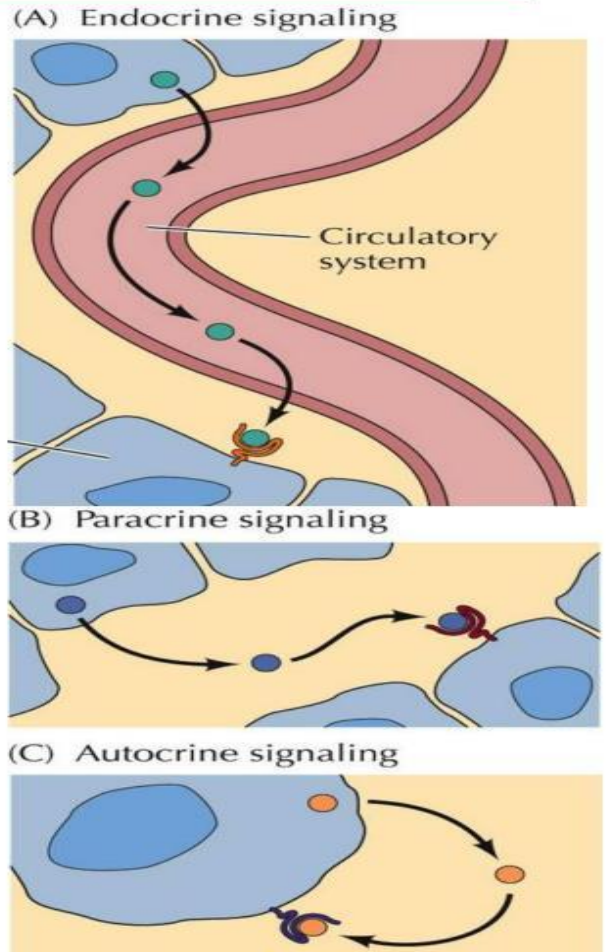
5 Types of Intercellular Signaling

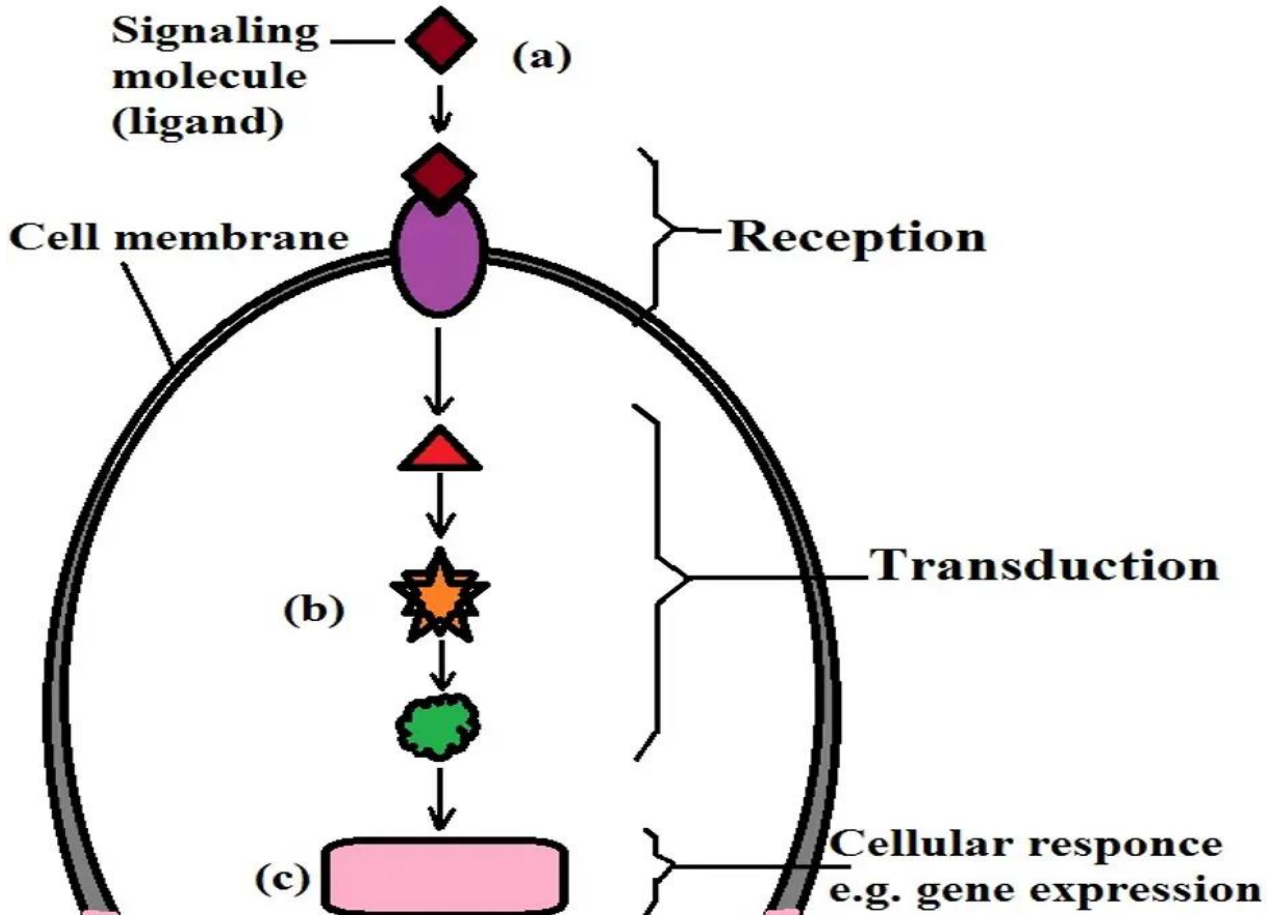
1. Endocrine signaling
2. Paracrine signaling
3. Synaptic signaling
4. Autocrine signaling
5. Signaling by plasma membrane-attached proteins



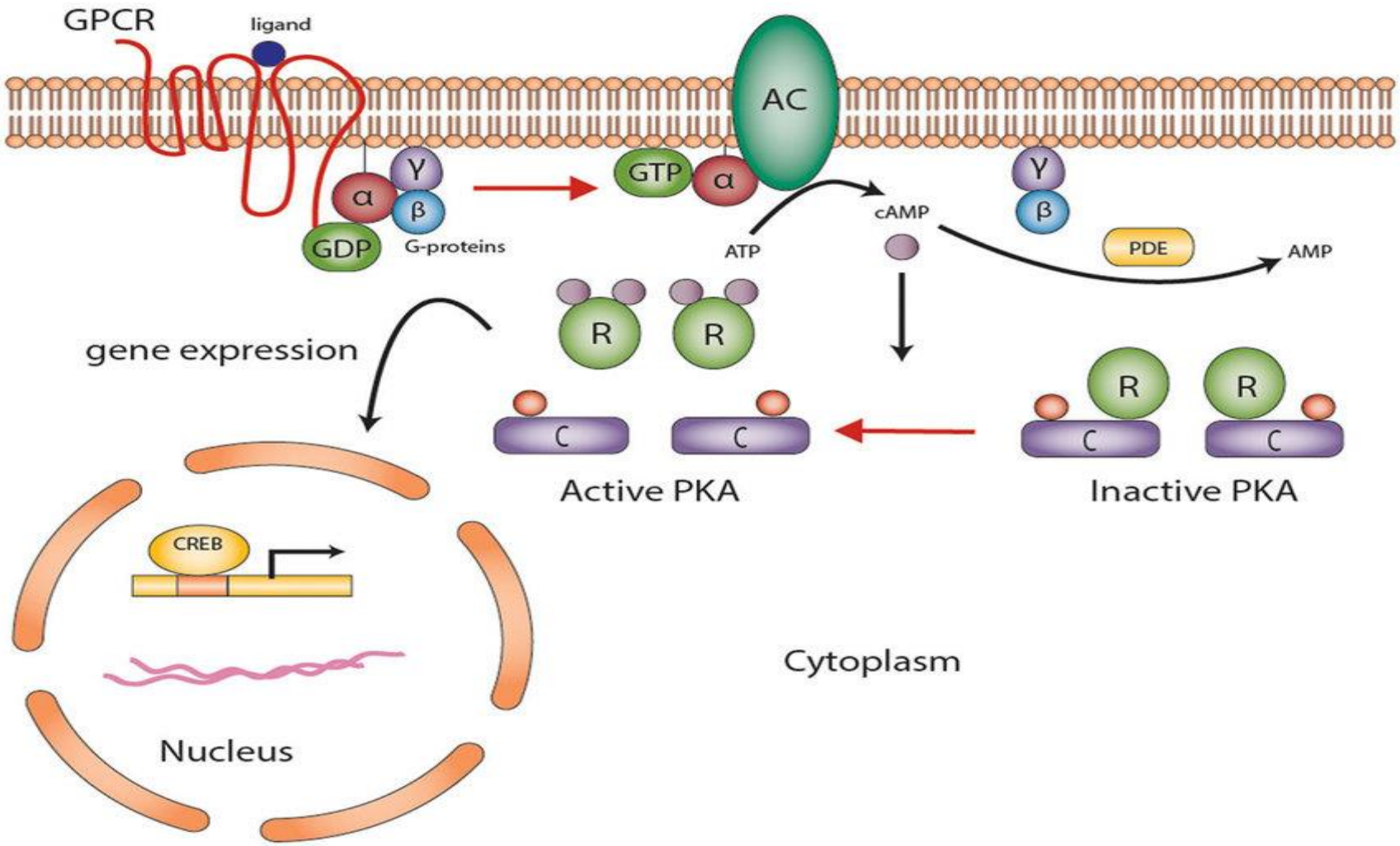
Signaling by secreted molecules

- **Endocrine signaling**—signaling molecules (**hormones**), target cells at distant body sites Example: estrogen.
- **Paracrine signaling**—act on neighboring target cells, e.g., neurotransmitters
- **Autocrine signaling**—respond to self signaling molecules
  - T lymphocytes: proliferation in response to cytokines produced by the same cell



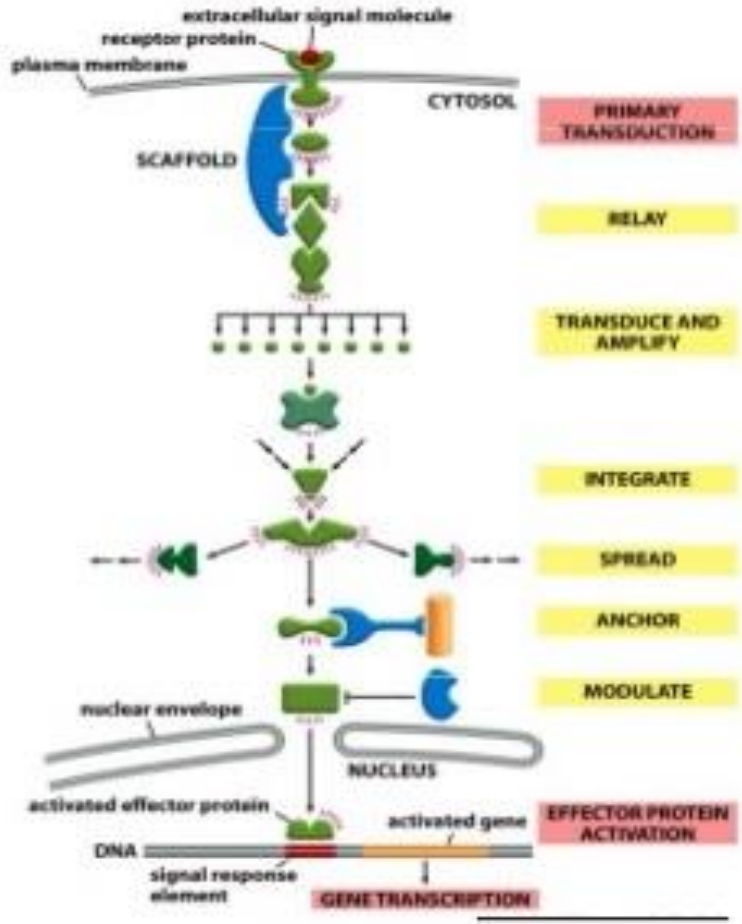

Signaling molecule (ligand)
(a)
Cell membrane
Reception
Transduction
(b)
(c)
Cellular responce
e.g. gene expression

G- protein signaling

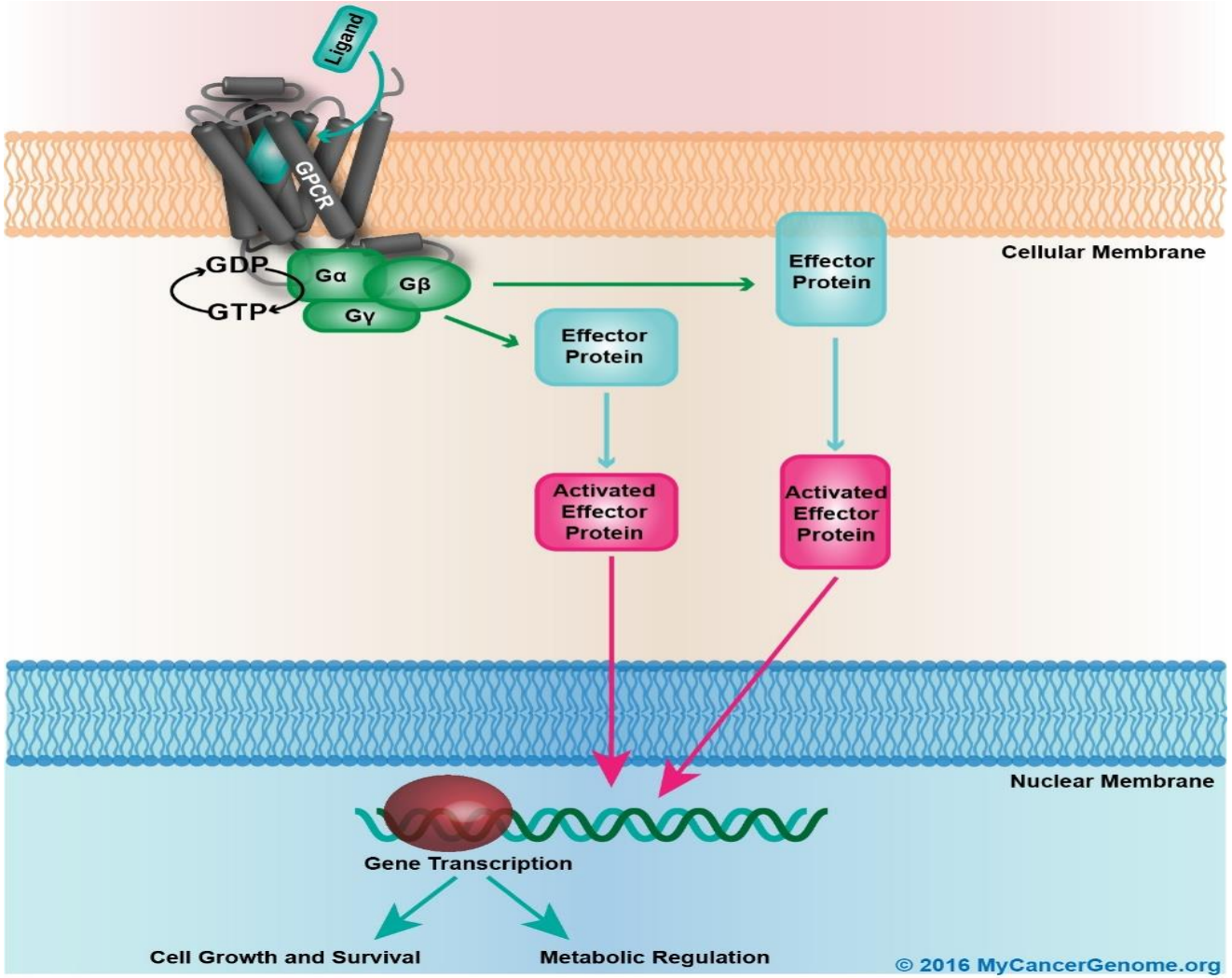


Overview of Cell Signaling

- Cascade of signaling events
  - ↓Receptors
  - ↓Signal transducers
  - ↓Effector proteins
- Relay, integration, and distribution of signals require transducers.
- Signaling pathways regulate nearly all cellular functions.

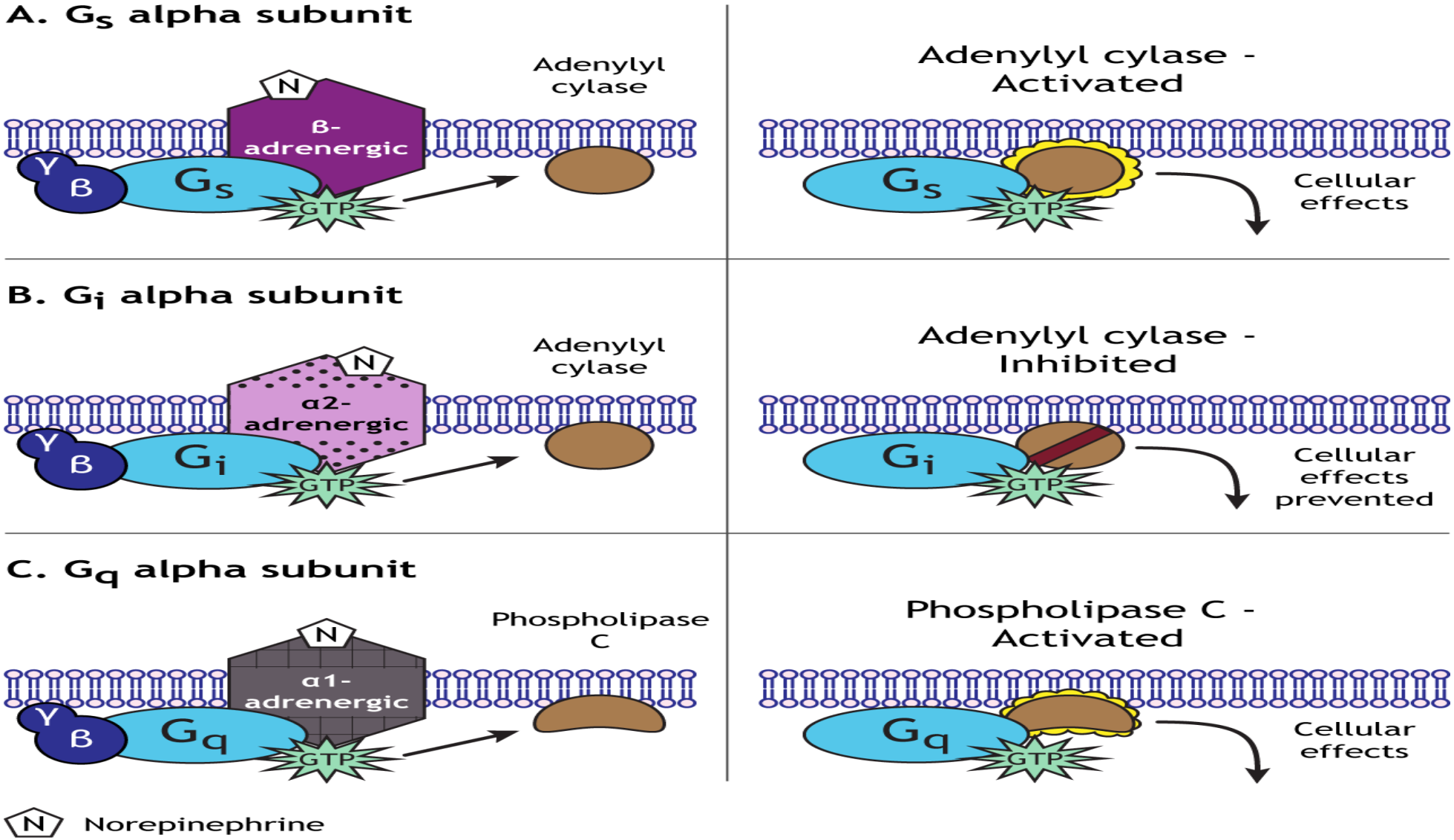


Conceptual Model of cell signaling



G–protein signalling

- norepinephrine can act on either alpha- or beta-adrenergic receptors.
- **Beta-adrenergic** GPCRs couple to a stimulatory G-protein, or G_s, which initiates the cyclic AMP (cAMP) second messenger system by activating the enzyme adenylyl cyclase.
- **Alpha 2-adrenergic** receptors, however, couple to an inhibitory G-protein, or G_i, and suppress the activity of adenylyl cyclase.
- **Alpha 1-adrenergic** receptors couple to a third type of G-protein, G_q, which activates the phospholipase C pathway.
- One neurotransmitter can, therefore, cause a wide range of cellular effects after binding to GPCRs, unlike the single function of ion flow through the ionotropic receptors.
- The pathway initiated by norepinephrine will depend on the type of receptor a specific cell expresses.


A. Gs alpha subunit
Adenylyl cylase
N
β-adrenergic
γ
β
Gs
GTP
Adenylyl cylase - Activated
Gs
GTP
Cellular effects
B. Gi alpha subunit
Adenylyl cylase
N
α2-adrenergic
γ
β
Gi
GTP
Adenylyl cylase - Inhibited
Gi
GTP
Cellular effects prevented
C. Gq alpha subunit
Phospholipase C
N
α1-adrenergic
γ
β
Gq
GTP
Phospholipase C - Activated
Gq
GTP
Cellular effects
N
Norepinephrine